How-To Library

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Configuring Advanced Properties for Data Domain Discovery in Test Data Management

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Abstract

You can edit the advanced properties of the Data Integration Service to improve the performance of a data domain profile that runs in Test Data Management (TDM). This article describes how to configure the advanced properties from Informatica Administrator. This article assumes that you have knowledge of the data discovery feature in TDM and are familiar with the Administrator tool.

Supported Versions

- Test Data Management 9.5.1 to 9.6.0

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Overview

You can create and run profiles to discover potential primary and foreign keys, entities, or data domains in TDM. Create a data domain to describe the columns you need to mask with the same data masking rules. When you create a data domain, you configure regular expressions that define patterns in the data or patterns in the column names.

When you run data domain discovery, the Data Integration Service processes all the column values to find matching inference for each data domain. The profiling services limit the number of data domains for each profile mapping because the complexity of each domain rule can vary. More data domains for each profile mapping can affect the processing resources of the Data Integration Service machine. When you run data domain discovery, the profiling services analyze the data source and profile parameters to determine the required set of mappings. The profiling services first categorize the volume of data in the data source and then generate the required number of mappings. You can edit the number of mappings and concurrent profiles to improve the profiling performance in TDM.

You cannot configure the number of mappings or profiles to be scheduled from Test Data Manager. You can configure settings from the Administrator tool which controls the profile execution. Configure the advanced properties of the Data Integration Service to configure the number of profile jobs run in parallel.

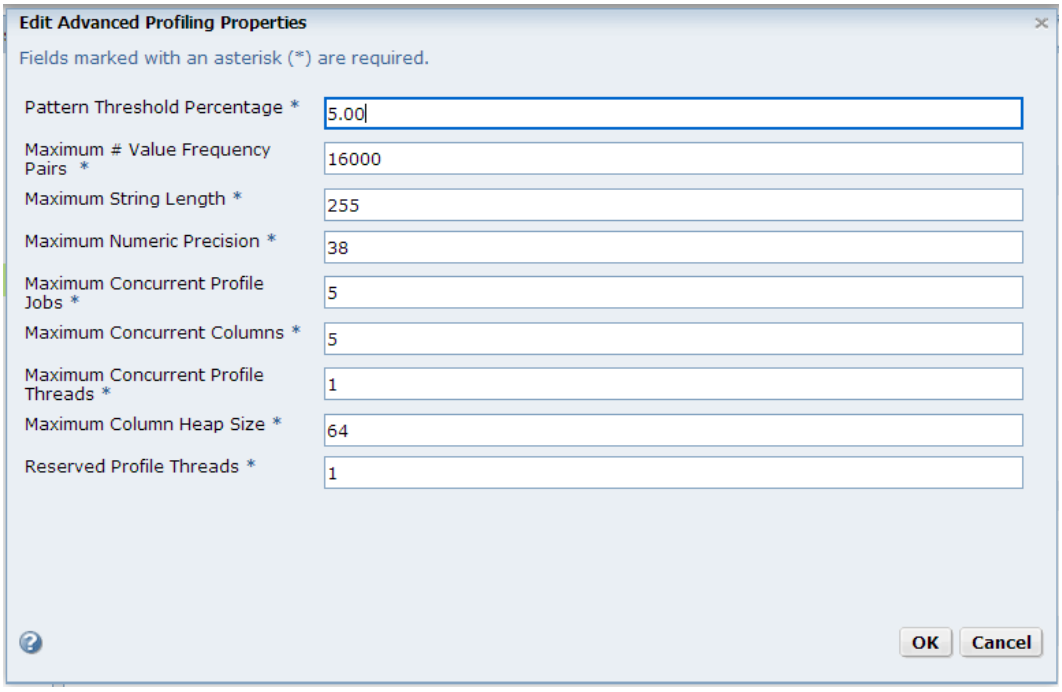
Editing Advanced Profiling Properties

You can edit the advanced properties of the Data Integration Service from the Administrator tool. You must have the required permission to log in to the Administrator tool.

To edit the advanced properties of the Data Integration Service, perform the following steps:

1. Log in to the Administrator tool.
2. From the **Domain Navigator**, select the Data Integration Service used for profiling.
3. In the **Properties** view, scroll down to the **Advanced Profiling Properties** section and click **Edit Advanced Profiling Properties**.

The following image shows the **Edit Advanced Profiling Properties** dialog box for the Data Integration Service in the Administrator tool:



4. Edit the following properties:

Maximum Concurrent Profile Jobs

The number of profile jobs that can run in parallel, even if there are more threads available to run mappings. You can use the property to control the number of concurrent jobs. Use the property to optimize the resources of the profiling services.

When you run a column profile on a relational source, the Maximum DB Connections property determines the number of mappings that the profiling services use. The profiling services use one mapping each for the other profiling jobs. When you run a column profile on flat file sources, the Maximum Concurrent Profile Threads property determines the number of mappings that the profiling services use.

You can configure the Maximum Concurrent Profile Jobs based on the capabilities and other uses of the nodes that you run the profiling services on.

Default is 5.

Maximum Concurrent Columns

The number of columns that a mapping runs in parallel. The default value of 5 is optimal for most of the profiling use cases. You can increase the default value for columns with cardinality lower than the average value. Decrease the default value for columns with cardinality higher than the average value. You might also want to decrease this value when you consistently run profiles on large source files where temporary disk space is low.

Default is 5.

Maximum Concurrent Profile Threads

The number of mappings that run in parallel when you run a column profile on a data source. Each mapping simultaneously runs a profile on a number of columns equal to the value you set for the Maximum Concurrent Columns property. If you increase this property value, the profiling services simultaneously run the profile on more columns and reduce the overall time for the profile run.

Default is 1.

Note: Setting the service properties can increase performance based on the CPU cores available. Configure the properties based on the CPU maximum and minimum threads suggested.

5. Optional. Edit the other properties. The following table describes the additional properties that you can edit:

Pattern Threshold Percent

The minimum percentage of rows matching up to two decimal places for a pattern to appear in the results.

Default is 5.00.

Maximum # Value Frequency Pairs

The maximum number of value frequency pairs stored in the profiling warehouse. This property does not control whether the profiling services compute all the value frequency pairs and the basic characteristics of a column profile run. However, an increase in the property value adds additional time to write more value frequency pairs to the profiling warehouse.

Default is 16,000.

Maximum String Length

The maximum length of a string that the profiling services mappings process internally. The default is set to the maximum value of 255. If you decrease the value, the Data Integration Service truncates the value. Decreased string lengths can have a minor impact on the amount of tablespace required for the profiling warehouse and negligible impact on the overall performance.

Default is 255.

Maximum Numeric Precision

The maximum precision, which is the number of significant digits in the number, for numeric decimal data types. If you set a low value for this property, the Data Integration Service might process additional numeric data types as strings instead of numbers.

Default is 38.

Maximum Column Heap Size

The cache size for each column profile mapping for flat files. You can increase this value to prevent the Data Integration Service from writing some parts of the intermediate profile results to temporary disk. However, this effect does not apply to large data sources. The default setting is optimal for most of the profiling use cases.

Default is 64.

Reserved Profile Threads

The number of threads that the Data Integration Service reserves to perform drill-down operations. The property ensures that a thread is always available for a drill-down operation, which is a quick and real-time operation. If enterprise discovery is a large part of the profile jobs, you can increase the property value. You can also increase the property value if multiple users might perform drill-down operations when the profiling services run a profile.

Default is 1.

6. To save the changes, click **OK**.
7. Restart the service.

Note: Log messages are generated for each profile mapping. If you select an entire table for the profile run, multiple log messages are generated for the profile.

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